

1 OSPF 基础实验

1.1 OSPF 的邻接关系和 LSA

1.1.1 实验介绍

1.1.1.1 学习目标

- 阐明在多路访问网络中接入多台路由器时的邻居关系建立过程
- 控制 OSPF DR 的选举
- 描述 5 种类型的 LSA 的内容，以及它们的作用

1.1.1.2 实验组网介绍

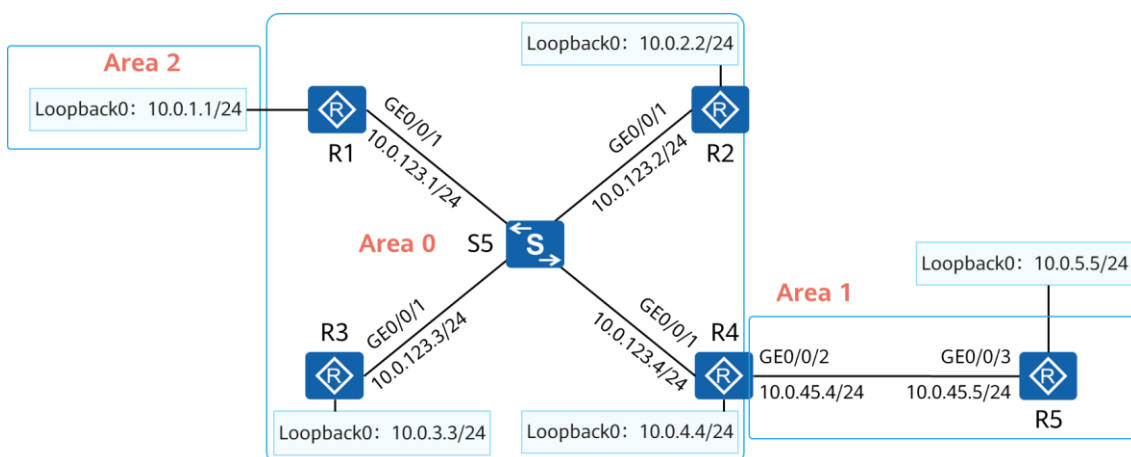


图1-1 OSPF 的邻接关系和 LSA

设备互联方式及 IP 地址规划如图所示，其中 R1 的 Loopback0 接口属于 OSPF 区域 2，R4 的 GE0/0/2 接口属于 OSPF 区域 1，R1、R2、R3、R4 的其他接口都属于 OSPF 区域 0。

R5 的 GE0/0/3 属于 OSPF 区域 1，R5 的 Loopback0 不属于 OSPF 域内。

1.1.1.3 实验背景

你是公司的网络管理员。现在公司的网络中有五台 AR 路由器，其中 R1、R2、R3 和 R4 在公司总部，通过以太网互联。R5 在公司分部，与公司总部的 R4 相连。由于网络规模较大，为了控制 LSA 的洪泛，你设计了多区域的 OSPF。

同时为了明确设备的 Router ID，你配置设备使用固定的地址作为 Router ID。

在 R1、R2、R3 与 R4 之间互联的网络上，需要干预 DR 与 BDR 的选举。实际使用中将 R3 定义为 DR、R2 定义为 BDR，R1、R4 定义为 DROther。

1.1.2 实验任务

1.1.2.1 任务思路

1. 设备 IP 地址配置。
2. 按照规划配置 OSPF 多区域。
3. 检查 OSPF 配置结果，检查 OSPF 邻居关系状态，检查 OSPF 路由表，检查 OSPF LSDB。
4. 手动修改接口的 DR 优先级，人工干预 OSPF DR、BDR 的选举结果。
5. 在 R5 上将直连路由引入到 OSPF 中，在 R1 上观察 Type-5 LSA。
6. 单独观察 Type-1 LSA、Type-2 LSA、Type-3 LSA、Type-4 LSA
7. R1 上通过 debug 观察 OSPF LSU、LSAck、LSR 报文。

1.1.2.2 任务步骤

步骤 1 互联接口、环回口 IP 地址配置

#设备命名

略

#关闭本实验中未使用的接口

略

#配置 R1 的 GE0/0/1、Loopback0 接口 IP 地址

```
[R1]interface GigabitEthernet0/0/1
[R1-GigabitEthernet0/0/1] ip address 10.0.123.1 24
[R1-GigabitEthernet0/0/1] quit
[R1]interface LoopBack 0
[R1-LoopBack0] ip address 10.0.1.1 24
[R1-LoopBack0] quit
```

#配置 R2 的 GE0/0/1、Loopback0 接口 IP 地址

```
[R2]interface GigabitEthernet0/0/1
[R2-GigabitEthernet0/0/1] ip address 10.0.123.2 24
[R2-GigabitEthernet0/0/1] quit
[R2]interface LoopBack 0
[R2-LoopBack0] ip address 10.0.2.2 24
[R2-LoopBack0] quit
```

#配置 R3 的 GE0/0/1、Loopback0 接口 IP 地址

```
[R3]interface GigabitEthernet0/0/1
[R3-GigabitEthernet0/0/1] ip address 10.0.123.3 24
[R3-GigabitEthernet0/0/1] quit
[R3]interface LoopBack 0
[R3-LoopBack0] ip address 10.0.3.3 24
[R3-LoopBack0] quit
```

#配置 R4 的 GE0/0/1、GE0/0/2、Loopback0 接口 IP 地址

```
[R4]interface GigabitEthernet0/0/1
[R4-GigabitEthernet0/0/1] ip address 10.0.123.4 24
[R4-GigabitEthernet0/0/1] quit
[R4] interface GigabitEthernet0/0/2
[R4-GigabitEthernet0/0/2] ip address 10.0.45.4 24
[R4-GigabitEthernet0/0/2] quit
[R4]interface LoopBack 0
[R4-LoopBack0] ip address 10.0.4.4 24
[R4-LoopBack0] quit
```

#配置 R5 的 GE0/0/3、Loopback0 接口 IP 地址

```
[R5]interface GigabitEthernet0/0/3
[R5-GigabitEthernet0/0/3] ip address 10.0.45.5 24
[R5-GigabitEthernet0/0/3] quit
[R5]interface LoopBack 0
[R5-LoopBack0] ip address 10.0.5.5 24
[R5-LoopBack0] quit
```

#在 R4 上检测互联地址连通性

```
<R4>ping -c 1 10.0.123.1
  PING 10.0.123.1: 56 data bytes, press CTRL_C to break
    Reply from 10.0.123.1: bytes=56 Sequence=1 ttl=255 time=100 ms

  --- 10.0.123.1 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 100/100/100 ms

<R4>ping -c 1 10.0.123.2
  PING 10.0.123.2: 56 data bytes, press CTRL_C to break
    Reply from 10.0.123.2: bytes=56 Sequence=1 ttl=255 time=110 ms

  --- 10.0.123.2 ping statistics ---
    1 packet(s) transmitted
    1 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 110/110/110 ms

<R4>ping -c 1 10.0.123.3
  PING 10.0.123.3: 56 data bytes, press CTRL_C to break
    Reply from 10.0.123.3: bytes=56 Sequence=1 ttl=255 time=110 ms
```

```
--- 10.0.123.3 ping statistics ---
  1 packet(s) transmitted
  1 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 110/110/110 ms

<R4>ping -c 1 10.0.45.5
  PING 10.0.45.5: 56 data bytes, press CTRL_C to break
    Reply from 10.0.45.5: bytes=56 Sequence=1 ttl=255 time=60 ms

--- 10.0.45.5 ping statistics ---
  1 packet(s) transmitted
  1 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 60/60/60 ms
```

步骤 2 配置多区域 OSPF

按照规划配置多区域 OSPF，修改 Loopback0 接口的网络类型为 Broadcast。

#配置 R1

```
[R1]ospf 1 router-id 10.0.1.1
[R1-ospf-1]area 0
[R1-ospf-1-area-0.0.0.0] network 10.0.123.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0] quit
[R1-ospf-1]area 2
[R1-ospf-1-area-0.0.0.2] network 10.0.1.1 0.0.0.0
[R1-ospf-1-area-0.0.0.2] quit
[R1-ospf-1]quit
[R1]interface LoopBack 0
[R1-LoopBack0] ospf network-type broadcast
[R1-LoopBack0] quit
```

#配置 R2

```
[R2]ospf 1 router-id 10.0.2.2
[R2-ospf-1]area 0
[R2-ospf-1-area-0.0.0.0] network 10.0.123.2 0.0.0.0
[R2-ospf-1-area-0.0.0.0] network 10.0.2.2 0.0.0.0
[R2-ospf-1-area-0.0.0.0] quit
[R2-ospf-1]quit
[R2]interface LoopBack 0
[R2-LoopBack0] ospf network-type broadcast
[R2-LoopBack0] quit
```

#配置 R3

```
[R3]ospf 1 router-id 10.0.3.3
[R3-ospf-1]area 0
[R3-ospf-1-area-0.0.0.0] network 10.0.123.3 0.0.0.0
[R3-ospf-1-area-0.0.0.0] network 10.0.3.3 0.0.0.0
[R3-ospf-1-area-0.0.0.0] quit
```

```
[R3-ospf-1]quit
[R3]interface LoopBack 0
[R3-LoopBack0] ospf network-type broadcast
[R3-LoopBack0] quit
```

#配置 R4

```
[R4]ospf 1 router-id 10.0.4.4
[R4-ospf-1]area 0
[R4-ospf-1-area-0.0.0.0] network 10.0.4.4 0.0.0.0
[R4-ospf-1-area-0.0.0.0] network 10.0.123.4 0.0.0.0
[R4-ospf-1-area-0.0.0.0] quit
[R4-ospf-1]area 1
[R4-ospf-1-area-0.0.0.1] network 10.0.45.4 0.0.0.0
[R4-ospf-1-area-0.0.0.1] quit
[R4-ospf-1]quit
[R4]interface LoopBack 0
[R4-LoopBack0] ospf network-type broadcast
[R4-LoopBack0] quit
```

#配置 R5

```
[R5]ospf 1 router-id 10.0.5.5
[R5-ospf-1]area 1
[R5-ospf-1-area-0.0.0.1] network 10.0.45.5 0.0.0.0
[R5-ospf-1-area-0.0.0.1] quit
[R5-ospf-1]quit
```

步骤 3 检查 OSPF 配置结果

#在 R4 上检查 OSPF 邻居的概要信息

```
[R4]display ospf peer brief
```

```
OSPF Process 1 with Router ID 10.0.4.4
Peer Statistic Information
```

Area Id	Interface	Neighbor id	State
0.0.0.0	GigabitEthernet0/0/1	10.0.1.1	Full
0.0.0.0	GigabitEthernet0/0/1	10.0.2.2	Full
0.0.0.0	GigabitEthernet0/0/1	10.0.3.3	2-Way
0.0.0.1	GigabitEthernet0/0/2	10.0.5.5	Full

R3、R4 之间只建立了邻居关系，而没有邻接关系。

#在 R4 上查看 OSPF 路由表

```
[R4]display ospf routing
```

```
OSPF Process 1 with Router ID 10.0.4.4
Routing Tables
```

```
Routing for Network
```

Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.4.0/24	0	Stub	10.0.4.4	10.0.4.4	0.0.0.0
10.0.45.0/24	1	Transit	10.0.45.4	10.0.4.4	0.0.0.1
10.0.123.0/24	1	Transit	10.0.123.4	10.0.4.4	0.0.0.0
10.0.1.0/24	1	Inter-area	10.0.123.1	10.0.1.1	0.0.0.0
10.0.2.0/24	1	Stub	10.0.123.2	10.0.2.2	0.0.0.0
10.0.3.0/24	1	Stub	10.0.123.3	10.0.3.3	0.0.0.0
Total Nets: 6					
Intra Area: 5 Inter Area: 1 ASE: 0 NSSA: 0					

#在 R5 上查看 OSPF LSDB

```
[R5]display ospflsdb
```

```
OSPFProcess 1 withRouter ID 10.0.5.5
Link State Database
```

Type	Area: 0.0.0.1	LinkState ID	AdvRouter	Age	Len	Sequence	Metric
Router		10.0.5.5	10.0.5.5	470	36	80000008	1
Router		10.0.4.4	10.0.4.4	1660	36	80000005	1
Network		10.0.45.4	10.0.4.4	1660	32	80000002	0
Sum-Net		10.0.3.0	10.0.4.4	1710	28	80000001	1
Sum-Net		10.0.2.0	10.0.4.4	1710	28	80000001	1
Sum-Net		10.0.1.0	10.0.4.4	1710	28	80000001	1
Sum-Net		10.0.4.0	10.0.4.4	1700	28	80000001	0
Sum-Net		10.0.123.0	10.0.4.4	1710	28	80000001	1

区域 1 中仅存在 2 台路由器，所以在 R5 的 LSDB 中，仅存在 2 条 Type-1 LSA，剩余的 5 条 Type-3 LSA 为 R4 向 R5 通告的区域间路由。

#在 R2 上查看 OSPF LSDB

```
[R2]display ospflsdb
```

```
OSPFProcess 1 withRouter ID 10.0.2.2
Link State Database
```

Type	Area: 0.0.0.0	LinkState ID	AdvRouter	Age	Len	Sequence	Metric
Router		10.0.3.3	10.0.3.3	256	48	8000000B	1
Router		10.0.4.4	10.0.4.4	211	48	8000000A	1
Router		10.0.2.2	10.0.2.2	268	48	8000000C	1
Router		10.0.1.1	10.0.1.1	270	36	8000000B	1
Network		10.0.123.1	10.0.1.1	270	40	80000007	0
Sum-Net		10.0.1.0	10.0.1.1	399	28	80000002	0
Sum-Net		10.0.45.0	10.0.4.4	265	28	80000002	1

在 R2 上除了 4 条 Type-1 LSA 以外，还有 1 条 Type-2 LSA。R2 的 GE 0/0/1 接口所连接的是一个广播型网络，该网络上的 DR 会产生 1 条 Type-2 LSA 来描述所有的邻居。在这里可以从 AdvRouter 字段得知生成这条 LSA 的路由器是 R1，符合 R1 是该网段 DR 的结果。

步骤 4 修改接口的 DR 优先级，影响 DR 选举

#修改 R4 的 GE0/0/1 接口的 DR 优先级为 255，确保 R4 成为 10.0.123.0/24 网段的 DR

```
[R4]interface GigabitEthernet0/0/1
[R4-GigabitEthernet0/0/1] ospf dr-priority 255
[R4-GigabitEthernet0/0/1] quit
```

#修改 R3 的 GE0/0/1 接口的 DR 优先级为 254，确保 R3 成为 10.0.123.0/24 网段的 BDR

```
[R3]interface GigabitEthernet0/0/1
[R3-GigabitEthernet0/0/1] ospf dr-priority 254
[R3-GigabitEthernet0/0/1] quit
```

#修改 R2 的 GE0/0/1 接口的 DR 优先级为 0，确保 R2 不参与选举

```
[R2]interface GigabitEthernet0/0/1
[R2-GigabitEthernet0/0/1] ospf dr-priority 0
[R2-GigabitEthernet0/0/1] quit
```

#为重新选举 DR、BDR，关闭再重新打开 R1、R2、R3、R4 的 GE0/0/1 接口

```
[R1]interface GigabitEthernet 0/0/1
[R1-GigabitEthernet0/0/1] shutdown

[R2]interface GigabitEthernet 0/0/1
[R2-GigabitEthernet0/0/1] shutdown

[R3]interface GigabitEthernet 0/0/1
[R3-GigabitEthernet0/0/1] shutdown

[R4]interface GigabitEthernet 0/0/1
[R4-GigabitEthernet0/0/1] shutdown

[R1]interface GigabitEthernet 0/0/1
[R1-GigabitEthernet0/0/1] undo shutdown
[R1-GigabitEthernet0/0/1] quit

[R2]interface GigabitEthernet 0/0/1
[R2-GigabitEthernet0/0/1] undo shutdown
[R2-GigabitEthernet0/0/1] quit

[R3]interface GigabitEthernet 0/0/1
[R3-GigabitEthernet0/0/1] undo shutdown
[R3-GigabitEthernet0/0/1] quit

[R4]interface GigabitEthernet 0/0/1
[R4-GigabitEthernet0/0/1] undo shutdown
[R4-GigabitEthernet0/0/1] quit
```

为保证选举结果按照设置的优先级决定，尽量同时重新打开接口，否则可能会出现先打开接口的路由器成为 DR、BDR。

#在 R3 上查看 DR、BDR 选举结果

```
<R3>display ospf peer
```

```
OSPF Process 1 with Router ID 10.0.3.3
Neighbors
```

```
Area 0.0.0.0 interface 10.0.123.3(GigabitEthernet0/0/1)'s neighbors
```

```
Router ID: 10.0.1.1      Address: 10.0.123.1
```

```
State: Full Mode:Nbr is Slave Priority: 1
```

```
DR: 10.0.123.4 BDR: 10.0.123.3 MTU: 0
```

```
Dead timer due in 40 sec
```

```
Retrans timer interval: 5
```

```
Neighbor is up for 00:59:26
```

```
Authentication Sequence: [ 0 ]
```

```
Router ID: 10.0.2.2      Address: 10.0.123.2
```

```
State: Full Mode:Nbr is Slave Priority: 0
```

```
DR: 10.0.123.4 BDR: 10.0.123.3 MTU: 0
```

```
Dead timer due in 36 sec
```

```
Retrans timer interval: 4
```

```
Neighbor is up for 00:59:36
```

```
Authentication Sequence: [ 0 ]
```

```
Router ID: 10.0.4.4      Address: 10.0.123.4
```

```
State: Full Mode:Nbr is Master Priority: 255
```

```
DR: 10.0.123.4 BDR: 10.0.123.3 MTU: 0
```

```
Dead timer due in 34 sec
```

```
Retrans timer interval: 0
```

```
Neighbor is up for 00:59:53
```

```
Authentication Sequence: [ 0 ]
```

此时 R4 为 DR、R3 为 BDR。

#在 R1 上查看 R1、R2 之间的邻居关系

```
<R1>display ospf peer brief
```

```
OSPF Process 1 with Router ID 10.0.1.1
Peer Statistic Information
```

Area Id	Interface	Neighbor id	State
0.0.0.0	GigabitEthernet0/0/1	10.0.2.2	2-Way
0.0.0.0	GigabitEthernet0/0/1	10.0.3.3	Full
0.0.0.0	GigabitEthernet0/0/1	10.0.4.4	Full

R1、R2 都为 DRother，它们之间保持在 2-Way 状态，只建立了邻居关系，不存在邻接关系。

步骤 5 将直连路由引入 OSPF 中

#R5 上 Loopback0 接口不属于 OSPF 区域，将 Loopback0 接口路由引入到 OSPF 中

```
[R5]ospf 1
```



```
[R5-ospf-1] import-route direct
```

#在 R1 上查看引入的外部路由条目

```
<R1>display ospf routing
```

```
OSPF Process 1 with Router ID 10.0.1.1
Routing Tables
```

```
Routing for Network
```

Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.1.0/24	0	Stub	10.0.1.1	10.0.1.1	0.0.0.2
10.0.123.0/24	1	Transit	10.0.123.1	10.0.1.1	0.0.0.0
10.0.2.0/24	1	Stub	10.0.123.2	10.0.2.2	0.0.0.0
10.0.3.0/24	1	Stub	10.0.123.3	10.0.3.3	0.0.0.0
10.0.4.0/24	1	Stub	10.0.123.4	10.0.4.4	0.0.0.0
10.0.45.0/24	2	Inter-area	10.0.123.4	10.0.4.4	0.0.0.0

```
Routing for ASEs
```

Destination	Cost	Type	Tag	NextHop	AdvRouter
10.0.5.0/24	1	Type2	1	10.0.123.4	10.0.5.5

```
Total Nets: 7
```

```
Intra Area: 5 Inter Area: 1 ASE: 1 NSSA: 0
```

Loopback0 接口路由已经成功被引入到 OSPF 中，以外部路由形式存在。

#在 R1 上查看 Type-5 LSA

```
<R1>display ospf lsdb ase
```

```
OSPF Process 1 with Router ID 10.0.1.1
Link State Database
```

```
Type      : External
Ls id     : 10.0.5.0
Adv rtr   : 10.0.5.5
Ls age    : 429
Len       : 36
Options   : E
seq#      : 80000001
chksum    : 0xa904
Net mask  : 255.255.255.0
TOS 0 Metric: 1
E type    : 2
Forwarding Address : 0.0.0.0
Tag       : 1
Priority   : Low
```

```
Type      : External
Ls id     : 10.0.45.0
Adv rtr   : 10.0.5.5
Ls age    : 429
Len       : 36
```

```
Options : E
seq# : 80000001
chksum : 0xef95
Net mask : 255.255.255.0
TOS 0 Metric: 1
E type : 2
Forwarding Address : 0.0.0.0
Tag : 1
Priority : Low
```

可以看到 Type-5 LSA 存在 2 条，但是 R1 的路由表中 OSPF 外部路由只存在 1 条 10.0.5.0/24，这是因为关于 10.0.45.0/24 路由同时还存在区域间路由，且区域间路由的优先级高于 OSPF 外部路由。

#在 R1 上查看 Type-3 LSA（以下只展示了区域 0 中的 Type-3 LSA）

```
<R1>display ospf lsdb summary
```

```
OSPF Process 1 with Router ID 10.0.1.1
Area: 0.0.0.0
Link State Database
```

```
Type : Sum-Net
Ls id : 10.0.1.0
Adv rtr : 10.0.1.1
Ls age : 1487
Len : 28
Options : E
seq# : 80000003
chksum : 0x72d1
Net mask : 255.255.255.0
Tos 0 metric: 0
Priority : Low
```

```
Type : Sum-Net
Ls id : 10.0.45.0
Adv rtr : 10.0.4.4
Ls age : 1506
Len : 28
Options : E
seq# : 80000003
chksum : 0x6fa1
Net mask : 255.255.255.0
Tos 0 metric: 1
Priority : Low
```

可以看到 Type-3 LSA 中同样存在 10.0.45.0/24，当 Type-3 LSA 和 Type-5 LSA 所描述的路由前缀及掩码相同时，OSPF 优选通过 Type-3 LSA 计算出的路由加载到路由表。

步骤 6 观察各种类型的 LSA

#在 R1 上查看 Type-1 LSA 10.0.1.1

```
<R1>display ospf lsdb router 10.0.1.1
```

```
OSPF Process 1 with Router ID 10.0.1.1
Area: 0.0.0.0
Link State Database
```

```
Type      : Router
Ls id     : 10.0.1.1
Adv rtr   : 10.0.1.1
Ls age    : 202
Len       : 36
Options   : ABR E
seq#      : 80000015
chksum    : 0x31e4
Link count: 1
```

```
* Link ID: 10.0.123.4
  Data    : 10.0.123.1
  Link Type: TransNet
  Metric : 1
```

```
Area: 0.0.0.2
```

```
Link State Database
```

```
Type      : Router
Ls id     : 10.0.1.1
Adv rtr   : 10.0.1.1
Ls age    : 180
Len       : 36
Options   : ABR E
seq#      : 80000005
chksum    : 0x1615
Link count: 1
```

```
* Link ID: 10.0.1.0
  Data    : 255.255.255.0
  Link Type: StubNet
  Metric : 0
  Priority : Low
```

对于 Type-1 LSA 来说，Ls id 字段表示生成这条 LSA 的路由器 Router ID。

从输出信息中可以看到 R1 生成了 2 条 Type-1 LSA，1 条在区域 0 中泛洪，1 条在区域 2 中泛洪。

在区域 0 中 R1 与一个 transmit 类型的网段相连，可以看到其 Link ID 字段的值为该网段上的 DR 接口 IP 地址，Data 部分为本地与 DR 相连的接口 IP 地址。

在区域 2 中，R1 的 Loopback0 接口属于该区域，可以看到其 Link Type 为 StubNet，其 Link ID 值为该 StubNet 网段的 IP 地址，Data 部分为 StubNet 网段的网络掩码。

#在 R2 上查看 Type-2 LSA

```
<R2>display ospf lsdb network
```

```
OSPF Process 1 with Router ID 10.0.2.2
```

```

Area: 0.0.0.0
Link State Database

Type       : Network
Ls id      : 10.0.123.4
Adv rtr    : 10.0.4.4
Ls age     : 817
Len        : 40
Options    : E
seq#       : 80000007
chksum     : 0x373d
Net mask   : 255.255.255.0
Priority    : Low
  Attached Router 10.0.4.4
  Attached Router 10.0.1.1
  Attached Router 10.0.2.2
  Attached Router 10.0.3.3

```

Type-2 LSA 由 DR 产生，从 Adv rtr 字段可以验证这一点（该 LSA 由 10.0.4.4 产生，即 DR 生成），对于 Type-2 LSA，其 Ls ID 字段值为该网段上 DR 的接口 IP 地址，Attached Router 为该网段上所有路由器的 Router ID。

#在 R1 上查看 Type-3 LSA 10.0.45.0

```
<R1>display ospf lsdb summary 10.0.45.0
```

```

OSPF Process 1 with Router ID 10.0.1.1
Area: 0.0.0.0
Link State Database

Type       : Sum-Net
Ls id      : 10.0.45.0
Adv rtr    : 10.0.4.4
Ls age     : 1290
Len        : 28
Options    : E
seq#       : 80000004
chksum     : 0x6da2
Net mask   : 255.255.255.0
Tos 0 metric: 1
Priority    : Low
Area: 0.0.0.2
Link State Database

```

```

Type       : Sum-Net
Ls id      : 10.0.45.0
Adv rtr    : 10.0.1.1
Ls age     : 1250
Len        : 28
Options    : E
seq#       : 80000004
chksum     : 0x9e76

```

```
Net mask : 255.255.255.0
Tos 0 metric: 2
Priority : Low
```

Type-3 LSA 的 Ls ID 字段值为网络前缀，而 net mask 携带了网络掩码，在 R1 上可以看到 2 条 Type-3 LSA。一条在区域 0 内，从 adv rtr 可以判断出为 R4 产生，由 R4 从区域 1 向区域 0 中通告产生，另外一条在区域 2 中，从 adv rtr 可以判断出由 R1 自身产生，R1 作为连接区域 0 与区域 2 的 ABR，同样会产生一条 Type-3 LSA，用于向区域 2 通告。

#在 R1 上查看 Type-4 LSA

```
<R1>display ospf lsdb asbr 10.0.5.5
```

```
OSPF Process 1 with Router ID 10.0.1.1
Area: 0.0.0.0
Link State Database
```

```
Type      : Sum-Asbr
Ls id      : 10.0.5.5
Adv rtr    : 10.0.4.4
Ls age     : 1257
Len        : 28
Options    : E
seq#       : 80000002
chksum     : 0xea49
Tos 0      metric: 1
Area: 0.0.0.2
Link State Database
```

```
Type      : Sum-Asbr
Ls id      : 10.0.5.5
Adv rtr    : 10.0.1.1
Ls age     : 1256
Len        : 28
Options    : E
seq#       : 80000002
chksum     : 0x1c1d
Tos 0      metric: 2
```

Type-4 LSA 用于描述如何到达 ASBR，从输出信息中可以看出 R1 上存在 2 条 Type-4 LSA，一条在区域 0 中，从 adv rtr 可知为 R4 产生，另外一条为 R1 自身产生，R1 作为连接区域 0 与区域 2 的 ABR，自身生成了一条 Type-4 LSA，其 Adv rtr 字段为自身 Router ID。

步骤 7 观察 LSR、LSU 和 LSAck

默认情况下，网络稳定运行，OSPF 路由器每 30 分钟更新一次。为了触发 OSPF 的查询和更新信息，取消在 R4 的 Loopback0 接口上激活 OSPF，在 R1 上观察 OSPF 报文。

#在 R1 上开启 **debugging ospf packet update**、**debugging ospf packet ack**

```
<R1>terminal debugging
Info: Current terminal debugging is on.
<R1>terminal monitor
Info: Current terminal monitor is on.
<R1>debugging ospf packet update
<R1>debugging ospf packet ack
```

#取消 R4 的 Loopback0 接口激活 OSPF 的配置

```
[R4]ospf 1
[R4-ospf-1]area 0
[R4-ospf-1-area-0.0.0.0] undo network 10.0.4.4 0.0.0.0
```

#查看 R1 的 debug 输出信息

```
May 25 2020 20:27:47.210.1-08:00 R1 RM/6/RMDEBUG:
FileID: 0x70178024 Line: 2218 Level: 0x20
OSPF 1: RECV Packet. Interface: GigabitEthernet0/0/1
May 25 2020 20:27:47.210.2-08:00 R1 RM/6/RMDEBUG: Source Address: 10.0.123.4
May 25 2020 20:27:47.210.3-08:00 R1 RM/6/RMDEBUG: Destination Address: 224.0.0.5
May 25 2020 20:27:47.210.4-08:00 R1 RM/6/RMDEBUG: Ver# 2, Type: 4 (Link-State Update)
May 25 2020 20:27:47.210.5-08:00 R1 RM/6/RMDEBUG: Length: 64, Router: 10.0.4.4
May 25 2020 20:27:47.210.6-08:00 R1 RM/6/RMDEBUG: Area: 0.0.0.0, Chksum: 5451
May 25 2020 20:27:47.210.7-08:00 R1 RM/6/RMDEBUG: AuType: 00
May 25 2020 20:27:47.210.8-08:00 R1 RM/6/RMDEBUG: Key(ascii): 0 0 0 0 0 0 0 0
May 25 2020 20:27:47.210.9-08:00 R1 RM/6/RMDEBUG: # LSAS: 1
May 25 2020 20:27:47.210.10-08:00 R1 RM/6/RMDEBUG: LSA Type 1
May 25 2020 20:27:47.210.11-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.4.4
May 25 2020 20:27:47.210.12-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.4.4
May 25 2020 20:27:47.210.13-08:00 R1 RM/6/RMDEBUG: LSA Age: 1
May 25 2020 20:27:47.210.14-08:00 R1 RM/6/RMDEBUG: Options: ExRouting:ON
May 25 2020 20:27:47.210.15-08:00 R1 RM/6/RMDEBUG: Length: 36, Seq# 80000017
May 25 2020 20:27:47.210.16-08:00 R1 RM/6/RMDEBUG: CheckSum: f014
May 25 2020 20:27:47.210.17-08:00 R1 RM/6/RMDEBUG: NtBit: 0 VBit: 0 EBit: 0 BBit: 1
May 25 2020 20:27:47.210.18-08:00 R1 RM/6/RMDEBUG: # Links: 1
May 25 2020 20:27:47.210.19-08:00 R1 RM/6/RMDEBUG: LinkID: 10.0.123.4
May 25 2020 20:27:47.210.20-08:00 R1 RM/6/RMDEBUG: LinkData: 10.0.123.4
May 25 2020 20:27:47.210.21-08:00 R1 RM/6/RMDEBUG: LinkType: 2
May 25 2020 20:27:47.210.22-08:00 R1 RM/6/RMDEBUG: TOS# 0 Metric 1
May 25 2020 20:27:47.210.23-08:00 R1 RM/6/RMDEBUG:
May 25 2020 20:27:47.570.1-08:00 R1 RM/6/RMDEBUG:
FileID: 0x70178024 Line: 2218 Level: 0x20
OSPF 1: RECV Packet. Interface: GigabitEthernet0/0/1
May 25 2020 20:27:47.570.2-08:00 R1 RM/6/RMDEBUG: Source Address: 10.0.123.3
May 25 2020 20:27:47.570.3-08:00 R1 RM/6/RMDEBUG: Destination Address: 224.0.0.5
May 25 2020 20:27:47.570.4-08:00 R1 RM/6/RMDEBUG: Ver# 2, Type: 5 (Link-State Ack)
May 25 2020 20:27:47.570.5-08:00 R1 RM/6/RMDEBUG: Length: 44, Router: 10.0.3.3
May 25 2020 20:27:47.570.6-08:00 R1 RM/6/RMDEBUG: Area: 0.0.0.0, Chksum: 6271
May 25 2020 20:27:47.570.7-08:00 R1 RM/6/RMDEBUG: AuType: 00
May 25 2020 20:27:47.570.8-08:00 R1 RM/6/RMDEBUG: Key(ascii): 0 0 0 0 0 0 0 0
May 25 2020 20:27:47.570.9-08:00 R1 RM/6/RMDEBUG: # LSA Headers: 1
May 25 2020 20:27:47.570.10-08:00 R1 RM/6/RMDEBUG: LSA Type 1
May 25 2020 20:27:47.570.11-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.4.4
May 25 2020 20:27:47.570.12-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.4.4
May 25 2020 20:27:47.570.13-08:00 R1 RM/6/RMDEBUG: LSA Age: 1
```

```

May 25 2020 20:27:47.570.14-08:00 R1 RM/6/RMDEBUG: Options: ExRouting:ON
May 25 2020 20:27:47.570.15-08:00 R1 RM/6/RMDEBUG: Length: 36, Seq# 80000017
May 25 2020 20:27:47.570.16-08:00 R1 RM/6/RMDEBUG: CheckSum: f014
May 25 2020 20:27:47.570.17-08:00 R1 RM/6/RMDEBUG:
May 25 2020 20:27:47.990.1-08:00 R1 RM/6/RMDEBUG:
FileID: 0x70178025 Line: 4427 Level: 0x20
OSPF 1: SEND Packet. Interface: GigabitEthernet0/0/1
May 25 2020 20:27:47.990.2-08:00 R1 RM/6/RMDEBUG: Source Address: 10.0.123.1
May 25 2020 20:27:47.990.3-08:00 R1 RM/6/RMDEBUG: Destination Address: 224.0.0.6
May 25 2020 20:27:47.990.4-08:00 R1 RM/6/RMDEBUG: Ver# 2, Type: 5 (Link-State Ack)
May 25 2020 20:27:47.990.5-08:00 R1 RM/6/RMDEBUG: Length: 44, Router: 10.0.1.1
May 25 2020 20:27:47.990.6-08:00 R1 RM/6/RMDEBUG: Area: 0.0.0.0, Chksum: 6472
May 25 2020 20:27:47.990.7-08:00 R1 RM/6/RMDEBUG: AuType: 00
May 25 2020 20:27:47.990.8-08:00 R1 RM/6/RMDEBUG: Key(ascii): 0 0 0 0 0 0 0 0
May 25 2020 20:27:47.990.9-08:00 R1 RM/6/RMDEBUG: # LSA Headers: 1
May 25 2020 20:27:47.990.10-08:00 R1 RM/6/RMDEBUG: LSA Type 1
May 25 2020 20:27:47.990.11-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.4.4
May 25 2020 20:27:47.990.12-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.4.4
May 25 2020 20:27:47.990.13-08:00 R1 RM/6/RMDEBUG: LSA Age: 2
May 25 2020 20:27:47.990.14-08:00 R1 RM/6/RMDEBUG: Options: ExRouting:ON
May 25 2020 20:27:47.990.15-08:00 R1 RM/6/RMDEBUG: Length: 36, Seq# 80000017
May 25 2020 20:27:47.990.16-08:00 R1 RM/6/RMDEBUG: CheckSum: f014

```

从输出信息中可以看到 3 个报文，第 1 个报文为：LSU，由 R4（DR）发送，目的地址 224.0.0.5，其中只包含 1 个网段，所以 Links 字段值为 1。

第二个报文 LSAck，由 R3（BDR）发送，目的地址 224.0.0.5。第 3 个报文同样为 LSAck，由 R1 发送，目的地址 224.0.0.6，该报文为 R1 发送给 DR、BDR。

#重新在 Loopback0 接口上激活 OSPF

```

[R4]ospf 1
[R4-ospf-1]area 0
[R4-ospf-1-area-0.0.0.0] network 10.0.4.4 0.0.0.0

```

#在 R1 上查看 debug 输出

```

May 25 2020 20:39:26.150.1-08:00 R1 RM/6/RMDEBUG:
FileID: 0x70178024 Line: 2218 Level: 0x20
OSPF 1: RECV Packet. Interface: GigabitEthernet0/0/1
May 25 2020 20:39:26.150.2-08:00 R1 RM/6/RMDEBUG: Source Address: 10.0.123.4
May 25 2020 20:39:26.150.3-08:00 R1 RM/6/RMDEBUG: Destination Address: 224.0.0.5
May 25 2020 20:39:26.150.4-08:00 R1 RM/6/RMDEBUG: Ver# 2, Type: 4 (Link-State Update)
May 25 2020 20:39:26.150.5-08:00 R1 RM/6/RMDEBUG: Length: 76, Router: 10.0.4.4
May 25 2020 20:39:26.150.6-08:00 R1 RM/6/RMDEBUG: Area: 0.0.0.0, Chksum: c8cf
May 25 2020 20:39:26.150.7-08:00 R1 RM/6/RMDEBUG: AuType: 00
May 25 2020 20:39:26.150.8-08:00 R1 RM/6/RMDEBUG: Key(ascii): 0 0 0 0 0 0 0 0
May 25 2020 20:39:26.150.9-08:00 R1 RM/6/RMDEBUG: # LSAS: 1
May 25 2020 20:39:26.150.10-08:00 R1 RM/6/RMDEBUG: LSA Type 1
May 25 2020 20:39:26.150.11-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.4.4
May 25 2020 20:39:26.150.12-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.4.4
May 25 2020 20:39:26.150.13-08:00 R1 RM/6/RMDEBUG: LSA Age: 2
May 25 2020 20:39:26.150.14-08:00 R1 RM/6/RMDEBUG: Options: ExRouting:ON
May 25 2020 20:39:26.150.15-08:00 R1 RM/6/RMDEBUG: Length: 48, Seq# 8000001b
May 25 2020 20:39:26.150.16-08:00 R1 RM/6/RMDEBUG: CheckSum: 6b77

```

```

May 25 2020 20:39:26.150.17-08:00 R1 RM/6/RMDEBUG: NtBit: 0 VBit: 0 EBit: 0 BBit: 1
May 25 2020 20:39:26.150.18-08:00 R1 RM/6/RMDEBUG: # Links: 2
May 25 2020 20:39:26.150.19-08:00 R1 RM/6/RMDEBUG: LinkID: 10.0.123.4
May 25 2020 20:39:26.150.20-08:00 R1 RM/6/RMDEBUG: LinkData: 10.0.123.4
May 25 2020 20:39:26.150.21-08:00 R1 RM/6/RMDEBUG: LinkType: 2
May 25 2020 20:39:26.150.22-08:00 R1 RM/6/RMDEBUG: TOS# 0 Metric 1
May 25 2020 20:39:26.150.23-08:00 R1 RM/6/RMDEBUG: LinkID: 10.0.4.0
May 25 2020 20:39:26.150.24-08:00 R1 RM/6/RMDEBUG: LinkData: 255.255.255.0
May 25 2020 20:39:26.150.25-08:00 R1 RM/6/RMDEBUG: LinkType: 3
May 25 2020 20:39:26.150.26-08:00 R1 RM/6/RMDEBUG: TOS# 0 Metric 0
May 25 2020 20:39:26.150.27-08:00 R1 RM/6/RMDEBUG:
May 25 2020 20:39:26.580.1-08:00 R1 RM/6/RMDEBUG:
FileID: 0x70178024 Line: 2218 Level: 0x20
OSPF 1: RECV Packet. Interface: GigabitEthernet0/0/1
May 25 2020 20:39:26.580.2-08:00 R1 RM/6/RMDEBUG: Source Address: 10.0.123.3
May 25 2020 20:39:26.580.3-08:00 R1 RM/6/RMDEBUG: Destination Address: 224.0.0.5
May 25 2020 20:39:26.580.4-08:00 R1 RM/6/RMDEBUG: Ver# 2, Type: 5 (Link-State Ack)
May 25 2020 20:39:26.580.5-08:00 R1 RM/6/RMDEBUG: Length: 44, Router: 10.0.3.3
May 25 2020 20:39:26.580.6-08:00 R1 RM/6/RMDEBUG: Area: 0.0.0.0, Chksum: e6fd
May 25 2020 20:39:26.580.7-08:00 R1 RM/6/RMDEBUG: AuType: 00
May 25 2020 20:39:26.580.8-08:00 R1 RM/6/RMDEBUG: Key(ascii): 0 0 0 0 0 0 0 0
May 25 2020 20:39:26.580.9-08:00 R1 RM/6/RMDEBUG: # LSA Headers: 1
May 25 2020 20:39:26.580.10-08:00 R1 RM/6/RMDEBUG: LSA Type 1
May 25 2020 20:39:26.580.11-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.4.4
May 25 2020 20:39:26.580.12-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.4.4
May 25 2020 20:39:26.580.13-08:00 R1 RM/6/RMDEBUG: LSA Age: 2
May 25 2020 20:39:26.580.14-08:00 R1 RM/6/RMDEBUG: Options: ExRouting:ON
May 25 2020 20:39:26.580.15-08:00 R1 RM/6/RMDEBUG: Length: 48, Seq# 8000001b
May 25 2020 20:39:26.580.16-08:00 R1 RM/6/RMDEBUG: CheckSum: 6b77
May 25 2020 20:39:26.580.17-08:00 R1 RM/6/RMDEBUG:
May 25 2020 20:39:26.910.1-08:00 R1 RM/6/RMDEBUG:
FileID: 0x70178025 Line: 4427 Level: 0x20
OSPF 1: SEND Packet. Interface: GigabitEthernet0/0/1
May 25 2020 20:39:26.910.2-08:00 R1 RM/6/RMDEBUG: Source Address: 10.0.123.1
May 25 2020 20:39:26.910.3-08:00 R1 RM/6/RMDEBUG: Destination Address: 224.0.0.6
May 25 2020 20:39:26.910.4-08:00 R1 RM/6/RMDEBUG: Ver# 2, Type: 5 (Link-State Ack)
May 25 2020 20:39:26.910.5-08:00 R1 RM/6/RMDEBUG: Length: 44, Router: 10.0.1.1
May 25 2020 20:39:26.910.6-08:00 R1 RM/6/RMDEBUG: Area: 0.0.0.0, Chksum: e8fe
May 25 2020 20:39:26.910.7-08:00 R1 RM/6/RMDEBUG: AuType: 00
May 25 2020 20:39:26.910.8-08:00 R1 RM/6/RMDEBUG: Key(ascii): 0 0 0 0 0 0 0 0
May 25 2020 20:39:26.910.9-08:00 R1 RM/6/RMDEBUG: # LSA Headers: 1
May 25 2020 20:39:26.910.10-08:00 R1 RM/6/RMDEBUG: LSA Type 1
May 25 2020 20:39:26.910.11-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.4.4
May 25 2020 20:39:26.910.12-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.4.4
May 25 2020 20:39:26.910.13-08:00 R1 RM/6/RMDEBUG: LSA Age: 3
May 25 2020 20:39:26.910.14-08:00 R1 RM/6/RMDEBUG: Options: ExRouting:ON
May 25 2020 20:39:26.910.15-08:00 R1 RM/6/RMDEBUG: Length: 48, Seq# 8000001b
May 25 2020 20:39:26.910.16-08:00 R1 RM/6/RMDEBUG: CheckSum: 6b77
May 25 2020 20:39:26.910.17-08:00 R1 RM/6/RMDEBUG:

```

首先第 1 个报文还是来自 R4 (DR) 的 LSU, 其中 Links 字段值为 2, 多了 1 个 Loopback0 接口路由, 第 2、第 3 个报文和之前一致: BDR 回应的 LSAck 以及 R1 自身回应的 LSAck。

#在 R1 上开启 **debugging ospf packet request**，重置 OSPF 进程

```
<R1>debugging ospf packet request
<R1>reset ospf process 1
```

#观察 R1 的 debug 输出

```
FileID: 0x70178025 Line: 2886 Level: 0x20
  OSPF 1: SEND Packet. Interface: GigabitEthernet0/0/1
May 25 2020 21:18:01.400.2-08:00 R1 RM/6/RMDEBUG: Source Address: 10.0.123.1
May 25 2020 21:18:01.400.3-08:00 R1 RM/6/RMDEBUG: Destination Address: 10.0.123.3
May 25 2020 21:18:01.400.4-08:00 R1 RM/6/RMDEBUG: Ver# 2, Type: 3 (Link-State Req)
May 25 2020 21:18:01.400.5-08:00 R1 RM/6/RMDEBUG: Length: 108, Router: 10.0.1.1
May 25 2020 21:18:01.400.6-08:00 R1 RM/6/RMDEBUG: Area: 0.0.0.0, Chksum: e85a
May 25 2020 21:18:01.400.7-08:00 R1 RM/6/RMDEBUG: AuType: 00
May 25 2020 21:18:01.400.8-08:00 R1 RM/6/RMDEBUG: Key(ascii): 0 0 0 0 0 0 0 0
May 25 2020 21:18:01.400.9-08:00 R1 RM/6/RMDEBUG: # Requesting LSAs: 7
May 25 2020 21:18:01.400.10-08:00 R1 RM/6/RMDEBUG: LSA Type 1
May 25 2020 21:18:01.400.11-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.1.1
May 25 2020 21:18:01.400.12-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.1.1
May 25 2020 21:18:01.400.13-08:00 R1 RM/6/RMDEBUG: LSA Type 1
May 25 2020 21:18:01.400.14-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.2.2
May 25 2020 21:18:01.400.15-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.2.2
May 25 2020 21:18:01.400.16-08:00 R1 RM/6/RMDEBUG: LSA Type 3
May 25 2020 21:18:01.400.17-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.45.0
May 25 2020 21:18:01.400.18-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.4.4
May 25 2020 21:18:01.400.19-08:00 R1 RM/6/RMDEBUG: LSA Type 3
May 25 2020 21:18:01.400.20-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.1.0
May 25 2020 21:18:01.400.21-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.1.1
May 25 2020 21:18:01.400.22-08:00 R1 RM/6/RMDEBUG: LSA Type 4
May 25 2020 21:18:01.400.23-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.5.5
May 25 2020 21:18:01.400.24-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.4.4
May 25 2020 21:18:01.400.25-08:00 R1 RM/6/RMDEBUG: LSA Type 5
May 25 2020 21:18:01.400.26-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.5.0
May 25 2020 21:18:01.400.27-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.5.5
May 25 2020 21:18:01.400.28-08:00 R1 RM/6/RMDEBUG: LSA Type 5
May 25 2020 21:18:01.400.29-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.45.0
May 25 2020 21:18:01.400.30-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.5.5
May 25 2020 21:18:01.400.31-08:00 R1 RM/6/RMDEBUG:
May 25 2020 21:18:01.430.1-08:00 R1 RM/6/RMDEBUG:
  FileID: 0x70178025 Line: 2886 Level: 0x20
  OSPF 1: SEND Packet. Interface: GigabitEthernet0/0/1
May 25 2020 21:18:01.430.2-08:00 R1 RM/6/RMDEBUG: Source Address: 10.0.123.1
May 25 2020 21:18:01.430.3-08:00 R1 RM/6/RMDEBUG:
May 25 2020 21:18:01.430.4-08:00 R1 RM/6/RMDEBUG: Ver# 2, Type: 3 (Link-State Req)
May 25 2020 21:18:01.430.5-08:00 R1 RM/6/RMDEBUG: Length: 108, Router: 10.0.1.1
May 25 2020 21:18:01.430.6-08:00 R1 RM/6/RMDEBUG: Area: 0.0.0.0, Chksum: e85a
May 25 2020 21:18:01.430.7-08:00 R1 RM/6/RMDEBUG: AuType: 00
May 25 2020 21:18:01.430.8-08:00 R1 RM/6/RMDEBUG: Key(ascii): 0 0 0 0 0 0 0 0
May 25 2020 21:18:01.430.9-08:00 R1 RM/6/RMDEBUG: # Requesting LSAs: 7
May 25 2020 21:18:01.430.10-08:00 R1 RM/6/RMDEBUG: LSA Type 1
May 25 2020 21:18:01.430.11-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.1.1
May 25 2020 21:18:01.430.12-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.1.1
```

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May 25 2020 21:18:01.430.13-08:00 R1 RM/6/RMDEBUG: LSA Type 1
May 25 2020 21:18:01.430.14-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.2.2
May 25 2020 21:18:01.430.15-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.2.2
May 25 2020 21:18:01.430.16-08:00 R1 RM/6/RMDEBUG: LSA Type 3
May 25 2020 21:18:01.430.17-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.1.0
May 25 2020 21:18:01.430.18-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.1.1
May 25 2020 21:18:01.430.19-08:00 R1 RM/6/RMDEBUG: LSA Type 3
May 25 2020 21:18:01.430.20-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.45.0
May 25 2020 21:18:01.430.21-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.4.4
May 25 2020 21:18:01.430.22-08:00 R1 RM/6/RMDEBUG: LSA Type 4
May 25 2020 21:18:01.430.23-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.5.5
May 25 2020 21:18:01.430.24-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.4.4
May 25 2020 21:18:01.430.25-08:00 R1 RM/6/RMDEBUG: LSA Type 5
May 25 2020 21:18:01.430.26-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.5.0
May 25 2020 21:18:01.430.27-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.5.5
May 25 2020 21:18:01.430.28-08:00 R1 RM/6/RMDEBUG: LSA Type 5
May 25 2020 21:18:01.430.29-08:00 R1 RM/6/RMDEBUG: LS ID: 10.0.45.0
May 25 2020 21:18:01.430.30-08:00 R1 RM/6/RMDEBUG: Adv Rtr: 10.0.5.5
```

从输出可以看到 R1 向 R3 (BDR)、R4 (DR) 发送了 LS Request。